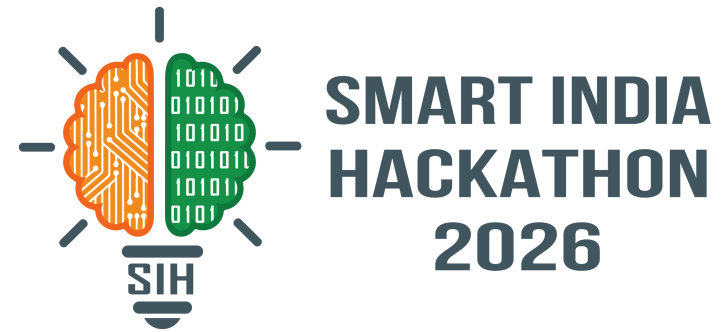
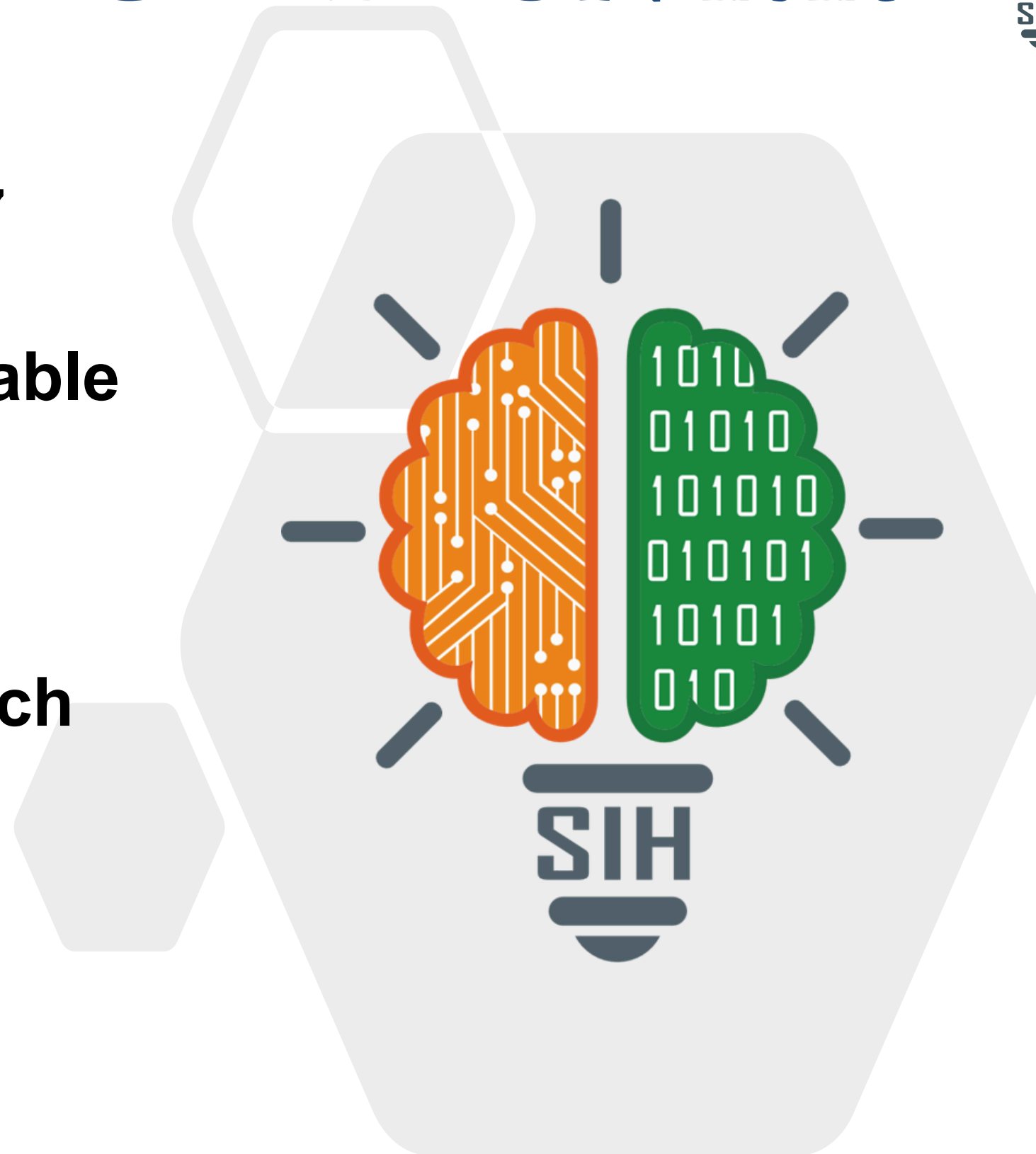


SMART INDIA HACKATHON 2026



- **Problem Statement ID – SIH26217**
- **Problem Statement Title- Renewable
/ Sustainable Energy**
- **Theme- Sustainability & Clean Tech**
- **PS Category- Hardware**
- **Team ID- GDT022**
- **Team Name- Symbiosis**



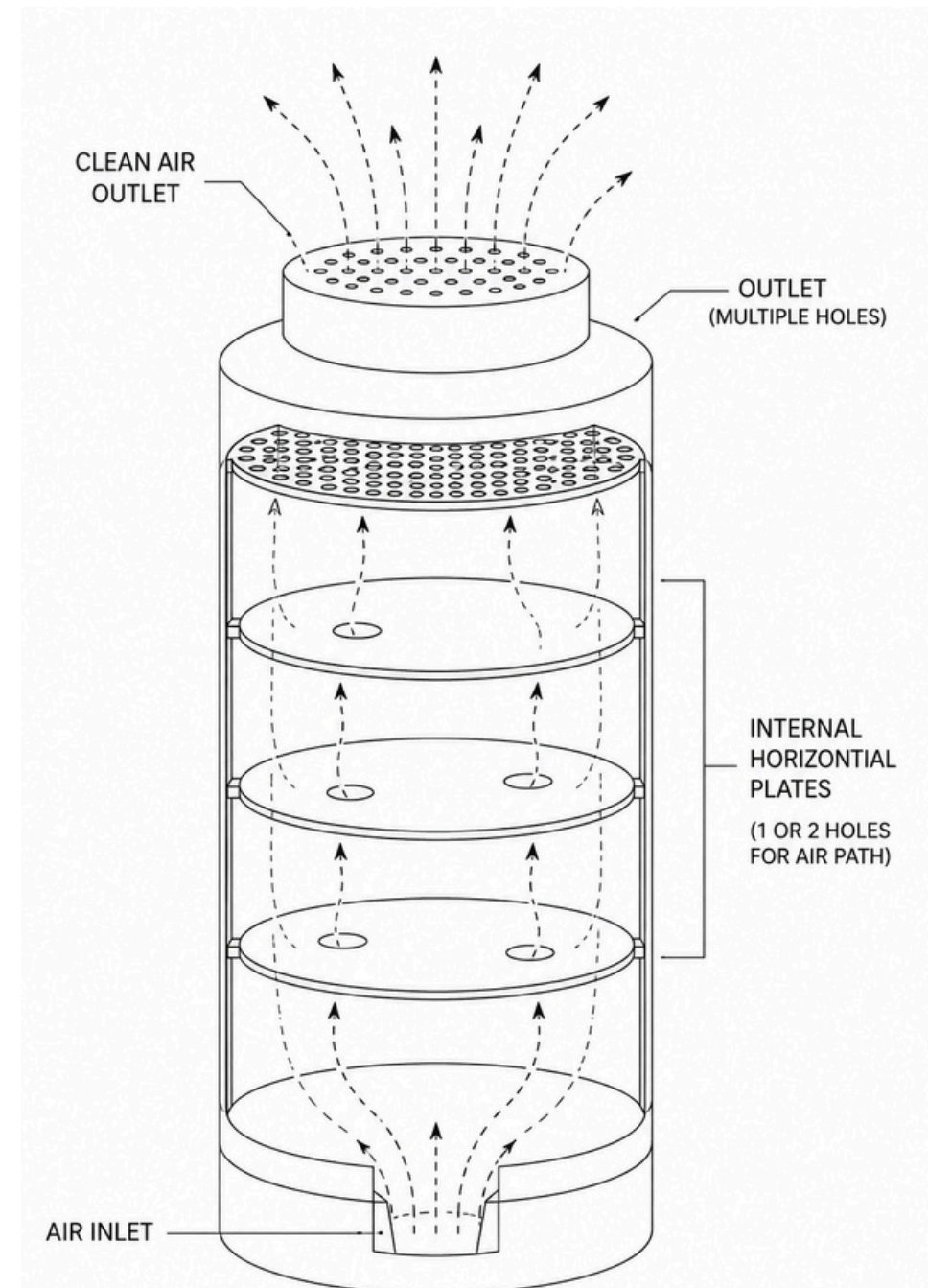
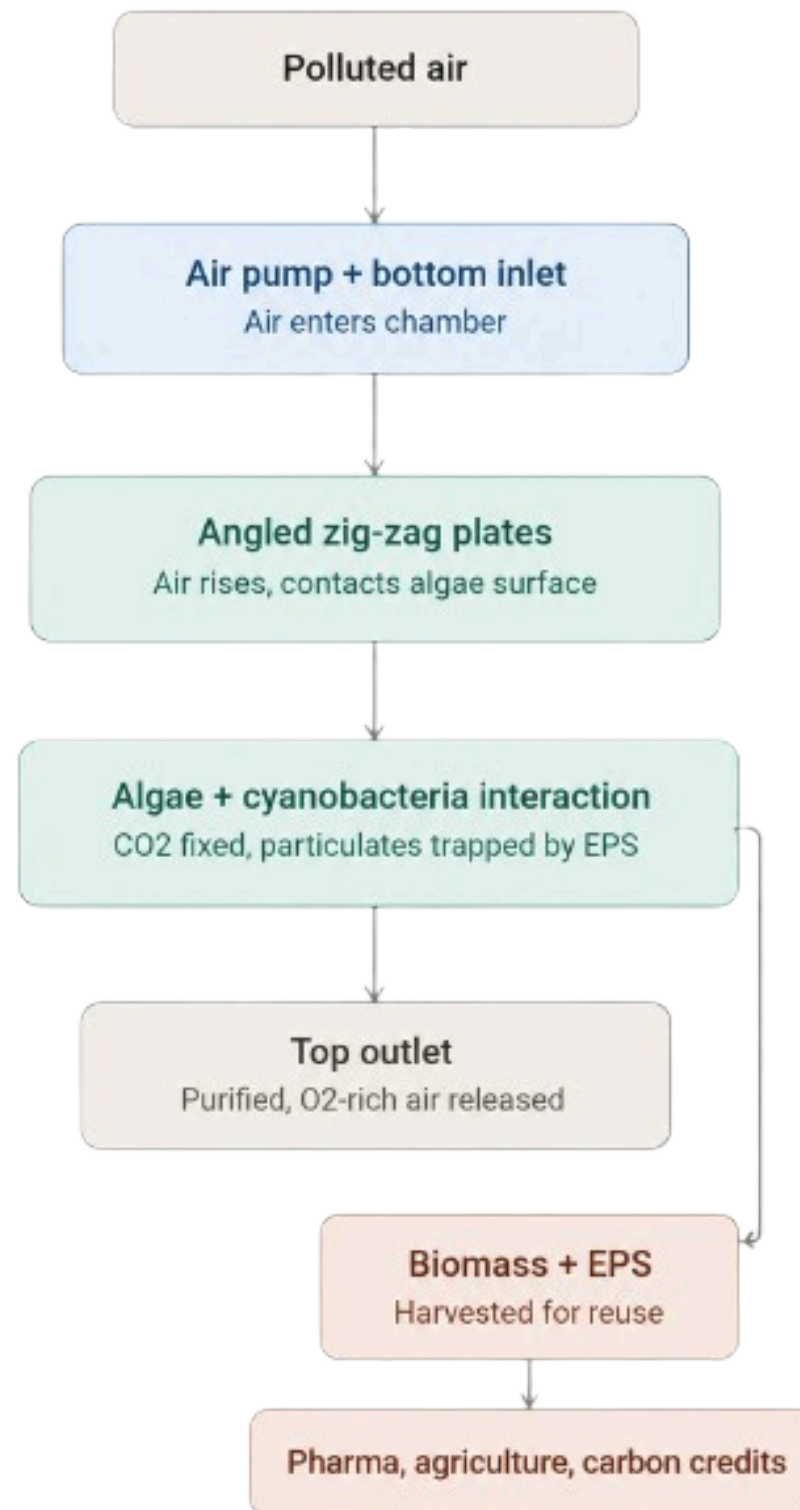
An Algae-Based Photobioreactor for Indoor Air Purification and CO₂ Mitigation



- Dual-algal photobioreactor: free-floating microalgae for CO₂ fixation + filamentous cyanobacteria for surface-based EPS particle trapping.
- Zig-zag bio-interaction plates (100°–105°) increase air–algae contact and residence time.
- Polluted indoor air enters through the bottom inlet, passes through multiple bioactive layers, and exits from the top as biologically treated, O₂-enriched air.
- Removable algae panels enable easy biomass/EPS harvesting and reuse



Indoor spaces face increasing PM_{2.5}/PM₁₀ and CO₂ accumulation, while conventional air purifiers mainly remove particulates, generate filter waste, and do not provide a sustainable solution for simultaneous pollutant removal and CO₂ mitigation.



Cylindrical geometry design for algae purifier system

- Air Quality Monitoring: PM_{2.5}/PM₁₀ ($\mu\text{g}/\text{m}^3$) & CO₂ (ppm).
- Algae Health Monitoring: pH, temperature ($^{\circ}\text{C}$), DO & turbidity
- Smart Data Acquisition: based real-time data collection & storage.
- Performance Optimization: Airflow(L/min) & light intensity (lux) monitoring

Feasibility

- **Technically:** Based on established algal cultivation and photobioreactor principles.
- **Cost-effective:** Uses low-cost, readily available components.
- **Scalable:** Modular design supports easy installation, maintenance and scale-up.
- **Sustainable:** Enables continuous CO₂ biofixation, O₂ generation and biomass production.

Challenges & Strategies

- Algal growth control: Use sensors and controlled LED lighting to maintain optimal growth conditions.
- Contamination & biofilm buildup: Follow regular cleaning, culture replacement and contamination-control protocols.
- Variable removal efficiency: Optimize airflow, algae density and plate angle through experimental testing.
- Maintenance requirements: Use removable modular panels for easy biomass harvesting and system cleaning.

- **CO₂ Fixation & O₂ Release:** Algae photosynthetically fix CO₂ and release O₂, supporting air-quality improvement.
- **PM & Pollutant Reduction:** Helps trap PM_{2.5}/PM₁₀ and remove nutrients and heavy metals such as Pb, Cd, Cr, Cu and Zn.
- **Wastewater Treatment:** Algal systems can significantly reduce N and P from wastewater.
- **Biomass Valorization:** Harvested biomass can be converted into EPS, algal fiber and biofertilizer.
- **Carbon Credits:** Verified CO₂ removal can potentially be converted into carbon credits (1 credit $\approx$ 1 tonne CO₂e).

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